Yuyuan Han

Computational imaging & physics-aware learning

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PhD student working on computational imaging and learning from optical measurements. Research experience spans measurement design, image reconstruction, and image-free perception.

EDUCATION & RESEARCH APPOINTMENTS

Xi'an Jiaotong University · PhD student — 2022 - present
Electronic Science and Technology · Master's-to-PhD track

Northwest University · B.Eng. — 2018 - 2022
Electronic Information Science and Technology

Politecnico di Milano · Visiting PhD student — Oct 2026 - present
Department of Physics · Host: Antonio Pifferi

SELECTED RESEARCH

The Lift Spectrum · Robust image-free perception — Preprint

arXiv · 2026 · Paper · Code

- Demonstrated greater robustness to measurement noise than the tested reconstruct-then-segment baselines, with all models trained on clean data.
- Learned spatial task representations directly from single-pixel measurements, using reconstruction only as auxiliary supervision.

Sparse Hadamard single-pixel imaging — Published

Optics Express · 2025 · Paper · Code

- Designed tunable-sparsity Hadamard illumination for noise-resilient single-pixel imaging; developed a noise model and validated the approach in optical experiments.

LDGI · Optical phased-array imaging — Published

Optics Communications · 2025 · Paper · Code

- Developed loop differential ghost imaging for line patterns from one-dimensional optical phased arrays; validated the algorithm in simulations and optical experiments.

CURRENT RESEARCH

Developing acquisition-aware models and adapting pretrained multimodal models to bucket-detector signals. Planned visiting research: time-domain diffuse optics and 3D reconstruction.

AWARD

2024 · Second Prize, National Post-Graduate Mathematical Contest in Modeling

ADDITIONAL EXPERIENCE

Photonic device modeling · Modeled and optimized electro-optic phased-array geometries for beam steering.